

SIMATS ENGINEERING

ASSESSMENT TOOL 1: Analytical Problem Solving

COURSE NAME: COMPUTER VISION

COURSE CODE: ITA05

COURSE OUTCOME COVERED

CO2: *Apply image enhancement and segmentation techniques for extracting meaningful regions from images. (BTL 3)*

Assessment Tool Weightage: 50%

QUESTIONS: 5

Total Marks 50

Duration :60 Minutes

Annexure A

Analytical Problem Solving

Code of Conduct:

I PALUKURU DHANUSH (192525141) certify that this submission is my original work and that I have adhered to the guidelines specified for this assessment. I understand that any violation of academic integrity rules will result in disciplinary action.

Signature of the student:

PALUKURU DHANUSH

RUBRICS FOR CALCULATION:

Criteria	Weightage	Level 4 (Exemplary)	Level 3 (Proficient)	Level 2 (Developing)	Level 1 (Beginning)	Marks (100)
Problem Understanding	20%	Clearly interprets problem, identifies all parameters and requirements accurately	Understands problem with minor gaps	Partial understanding; misses key elements	Misinterprets or unable to understand problem	
Method Selection	20%	Selects most appropriate method/formula with strong justification	Selects correct method with minor errors	Inappropriate or partially correct method	Unable to select method	
Solution Process	25%	Logical, systematic steps with correct approach throughout	Mostly correct steps with minor mistakes	Disorganized steps; multiple errors	No clear process	
Accuracy of Calculation	20%	All calculations accurate; correct final answer	Minor calculation errors	Frequent errors; incorrect final answer	No valid calculations	

Interpretation of Results	15%	Clearly interprets results with units and engineering relevance	Basic interpretation with minor omissions	Limited or unclear interpretation	No interpretation	
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1. Grey Level Transformation

(a) Explain the purpose of grey level transformation in image enhancement.

Ans: Grey level transformation is used to improve the visual quality of an image by changing the intensity values of pixels. It helps in:

- Enhancing image contrast.
- Highlighting important features.
- Brightening or darkening an image.
- Preparing images for further processing.

(b) Apply a negative transformation $s=255-r$ to the pixel values:

$r=[50,100,150,200]$

Compute the transformed values.

Ans: Formula:

$$s = 255 - r$$

Original Pixel (r)	Transformed Pixel (s)
50	205
100	155
150	105
200	55

Answer:

[205,155,105,55]

2. Log Transformation

(a) Explain how log transformation enhances low-intensity pixels.

Ans: Log transformation expands dark pixel values while compressing brighter ones. This makes details hidden in dark regions more visible.

Formula:

$$s = c \log(1 + r)$$

where $c = 10$.

(b) Given $s=c\log(1+r)$, where $c=10$, compute s for:

$r=[0,10,50,100]$

Ans: Using natural logarithm (ln):

For $r = 0$

$$s = 10 \ln(1) = 0$$

For $r = 10$

$$s = 10 \ln(11) = 10(2.398) = 23.98$$

For $r = 50$

$$s = 10\ln(51) = 10(3.932) = 39.32$$

For $r = 100$, $s = 10\ln(101) = 10(4.615) = 46.15$

r	s
0	0.00
10	23.98
50	39.32
100	46.15

Answer:[0,23.98,39.32,46.15]

3. Histogram Processing

(a) Explain the role of histogram equalization in contrast enhancement.

Ans: Histogram equalization improves image contrast by spreading the intensity values over the entire available range. It increases visibility of image details, especially in low-contrast images.

(b) Given a 3-bit image with histogram:

Intensity	Frequency
0	2
1	2
2	4
3	8

Compute the cumulative distribution and equalized intensity values.

Ans: Given histogram:

Intensity	Frequency
0	2
1	2
2	4
3	8

Total pixels

$$N = 2 + 2 + 4 + 8 = 16$$

Step 1: Cumulative Frequency

Intensity	Frequency	CDF
0	2	2
1	2	4
2	4	8
3	8	16

Step 2: Equalized Intensity

For a **3-bit image**, maximum intensity

$$L = 8$$

Formula

$$s = \frac{CDF}{N}(L - 1)$$

$$L - 1 = 7$$

Intensity	CDF	Equalized Value
0	2	$0.875 \approx 1$
1	4	$1.75 \approx 2$
2	8	$3.5 \approx 4$
3	16	7

Answer:

Original	Equalized
0	1
1	2
2	4
3	7

4. Spatial Filtering (Smoothing)

(a) Explain the purpose of spatial filtering for smoothing.

Ans: Spatial smoothing reduces image noise and removes small unwanted variations by replacing each pixel with the average of its neighboring pixels.

(b) Apply a 3×3 averaging filter to the following region:

10 20 30 20 30 40 30 40 50

Compute the new center pixel value.

Ans: **Apply 3×3 Averaging Filter**

Matrix

$$\begin{bmatrix} 10 & 20 & 30 \\ 20 & 30 & 40 \\ 30 & 40 & 50 \end{bmatrix}$$

Sum

$$10 + 20 + 30 + 20 + 30 + 40 + 30 + 40 + 50 = 270$$

Average

$$\frac{270}{9} = 30$$

New center pixel

30

5. Spatial Filtering (Sharpening)

(a) Explain how sharpening enhances image details.

Ans: Sharpening enhances edges and fine details by emphasizing rapid intensity changes. It improves image clarity and object boundaries.

(b) Apply the Laplacian mask:

$\begin{bmatrix} 0 & -1 & 0 \end{bmatrix}$, $\begin{bmatrix} -1 & 4 & -1 \end{bmatrix}$, $\begin{bmatrix} 0 & -1 & 0 \end{bmatrix}$

to the matrix:

10 10 10, 10 20 10, 10 10 10

Compute the output at the center.

Mask

$$\begin{bmatrix} 0 & -1 & 0 \\ -1 & 4 & -1 \\ 0 & -1 & 0 \end{bmatrix}$$

Image

$$\begin{bmatrix} 10 & 10 & 10 \\ 10 & 20 & 10 \\ 10 & 10 & 10 \end{bmatrix}$$

Calculation at center:

$$\begin{aligned} &(4 \times 20) - (10 + 10 + 10 + 10) \\ &80 - 40 = 40 \end{aligned}$$

Output at center

$$\boxed{40}$$